

Regular Monitoring and Detection Metrics Dashboard Design

Templates - Model: DASHBOARD-010

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1. Introduction

The **Regular Monitoring and Detection Metrics Dashboard Design** serves as a comprehensive template for security operation teams tasked with continuous threat detection and incident response. This document provides a detailed framework for building dashboards that enable real-time visibility into beaconing activity, authentication failures, suspicious process executions, network exfiltration attempts, and checkout anomalies.

Informed by threat-hunting playbooks and incident response logs from the past six months, this template incorporates specific detection signatures, EDR artifacts, IAM audit notes, network blocklists, and failure codes. The goal is to

facilitate early detection, rapid response, and effective containment of security threats.

2. Dashboard Objectives and Scope

The primary objectives of the monitoring dashboards include:

- **Early Warning:** Detect suspicious beaconing and command-and-control (C2) activities.
- **Failure Rate Monitoring:** Track Multi-Factor Authentication (MFA) failures and checkout errors to identify potential brute-force or automation attempts.
- **Unauthorized Process Detection:** Identify suspicious endpoint processes such as mshta.exe invoking PowerShell with encoded commands.
- **Network Exfiltration:** Flag anomalous network activity involving known malicious IPs or domains.
- **Incident Response Support:** Provide actionable insights for containment and remediation efforts.

The scope encompasses both endpoint activities and network behavior, integrating data from multiple security tools and logs. The dashboards are designed to support security analysts, incident responders, and threat hunters in maintaining situational awareness.

3. Detection Signatures and Artifacts

3.1 Threat Signatures and Identifiers

Based on recent threat-hunting activities, the following detection signatures and patterns are critical:

- **BEACON-12xx Series:** Indicators of beaconing behavior in network traffic, signaling compromised endpoints communicating with C2 servers.
- **C2-DNS-30xx Series:** DNS query patterns indicative of command and control traffic leveraging DNS tunneling or fast-flux domains.

3.2 EDR Artifacts and Hashes

Endpoint Detection and Response (EDR) tools register specific process artifacts, including:

- Suspicious process: `mshta.exe` executed with encoded PowerShell commands — often used for obfuscation.
- SHA256 Hashes of malicious or suspicious files:

```
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca49599
d41d8cd98f00b204e9800998ecf8427e
```

3.3 IAM Audit Events

Audit logs reveal suspicious identity and access anomalies, such as:

- **Auto-Creation Events:** Service accounts automatically created without proper authorization, e.g., *service account auto-creation*.
- **Scope IDs:** Unusual scope IDs like *SCOPE-PROD-API-ADMIN* or *SCOPE-AGENT-X* assigned during provisioning.
- **Login Anomalies:** Multiple failed login attempts or unusual source IPs.

3.4 Network Block List Indicators

Monitoring network blocks involves tracking known malicious domains and IP ranges, such as:

Type	Identifier	Description
Domain	c2-host-01.example	Known C2 server domain involved in recent campaigns
IP Range	192.168.100.0/24	Suspicious internal subnet communicating with external host
ASN	ASN-64512	Autonomous System Number linked to malicious activity

3.5 Checkout Failure Codes

Analysis of failure codes assists in identifying fraud or operational issues, including:

- **PAY-408:** Timeout during payment processing, possibly indicating automated attack attempts.
- **GATEWAY-52:** Gateway errors linked to suspicious transactions.
- **CART-ERR-1203:** Checkout errors caused by invalid session or attempted abuse.

4. Dashboard Design Guidelines

Creating effective dashboards for this monitoring requires adherence to several design principles:

- **Clarity and Readability:** Use clear labels, logical grouping, and distinct visual cues.
- **Real-time Data Refresh:** Enable live updates with minimal latency for timely detection.
- **Filtering and Drill-Downs:** Include filter options for date ranges, asset types, and specific signatures.
- **Alert Thresholds:** Visual markers (colors, badges) indicating thresholds crossed (e.g., MFA failure rates above 5%).
- **Historical Comparison:** Overlay current metrics with historical data to identify anomalies.

Sample visualization components include trend lines, bar charts for failure rates, heatmaps for suspicious activity, and tables for detailed event logs.

5. Core Monitoring Metrics

5.1 Beaconing and C2 Activity Metrics

- Number of beaconing events per hour
- Distinct C2 domains queried over time
- Frequency and timing patterns consistent with malware activity

5.2 Authentication Failure Rates

- MFA failure count per user/site

- Failure rate trend over specified timeframes
- Failed login attempts from unknown IP ranges

5.3 Suspicious Endpoint Processes

- Count and list of processes such as mshta.exe or PowerShell invoked with encoded commands
- Process invocation frequency patterns

5.4 Network Exfiltration Indicators

- Outbound traffic volume to known malicious IPs
- Number of connections to blocked domains or IPs
- Detection of data transfer anomalies (size, duration)

5.5 Checkout and Payment Failures

- Number of specific failure codes: PAY-408, GATEWAY-52, CART-ERR-1203
- Temporal trends in checkout failures
- Correlation with suspicious IPs or user behaviors

6. Examples of Detection Signatures and Alerts

6.1 Beaconing Detect Example

An alert generated when multiple beaconing events are observed from a single host to the same domain within a short interval (e.g., 5 minutes), matching signature **BEACON-1200**.

Pattern:

- More than 10 DNS requests to c2-host-01.example within 5 minutes
- Requests follow a periodic pattern (e.g., every 4 minutes)
- Associated with endpoint process svchost.exe or powershell.exe

6.2 MFA Failure Anomaly

Detection of an abnormal spike in MFA failures, e.g., exceeding a threshold of 5 failures in 1 minute for a particular user account, triggering an alert **AUTH-FAIL-005**.

6.3 Suspicious Process Execution

Alerts when `mshta.exe` launches with an encoded PowerShell command:

```
Command:
mshta.exe "vbscript:Execute(\"powershell -EncodedCommand [Base64]
Detection Signature:
Process name: mshta.exe
Command line includes encoded PowerShell commands, obfuscated
```

6.4 Penetration and Exfiltration

Outbound connections to IPs in the block list exceeding normal thresholds (e.g., >100MB data transfer in an hour), indicating possible exfiltration.

7. Incident Response and Containment Procedures

7.1 Containment Steps

1. Isolate affected endpoints detected with suspicious processes or beaconing activity.
2. Terminate malicious processes (e.g., `mshta.exe`, PowerShell with encoded commands).
3. Block malicious IP addresses and domains identified via network traffic analysis.
4. Reset compromised user credentials and enforce MFA where applicable.
5. Disable suspicious service accounts created without approval.

7.2 API Throttling Playbook

To prevent automated attack automation, the API throttling protocol involves:

- Implementing rate limits (e.g., 100 requests per minute per IP)
- Enforcing CAPTCHA challenges after threshold breaches
- Integrating with Web Application Firewalls for real-time blocking

7.3 Partner OAuth Review Procedures

Regular review of partner OAuth apps involves:

- Verifying scopes associated with each application (e.g., *SCOPE-PROD-API-ADMIN*)
- Auditing OAuth token issuance logs for anomalies
- Revoking unnecessary or suspicious OAuth client credentials

8. Appendices

8.1 References

- Threat-Hunting Playbook (Last 6 months).
- Incident Response Procedures Documentation.
- Security Analytics and SIEM Configuration Guides.
- Known attack signatures and indicators from threat intelligence feeds.

8.2 Glossary of Terms

Term	Definition
Beaconing	A pattern of periodic outbound network traffic, often used by malware to communicate with C2 servers.
C2	Command and Control server used by attackers to issue commands to compromised hosts.
IAM	Identity and Access Management, managing user identities and permissions.
EDR	Endpoint Detection and Response tools that monitor and analyze endpoint activity for threats.